

Shooter Performance Feature Engineering - VR Shooting Simulation

This notebook turns raw VR shooting logs (controller/headset/eye-tracking frames plus drill/repetition/shot events) into five interpretable shot-quality features, one per performance domain, plus a derived sixth (decision quality). Two more metrics were built and validated along the way but didn't get promoted to the main set; Section 2 covers why.

It then presents those features twice: as a plain-language dashboard for the shooter, and as a PM-facing writeup of why each one was chosen and how it could map to product feedback. See `CLAUDE.md` for the data dictionary and `README.md` for how to run this.

Cohort: 7 users, each running the same `TargetAcquisition` scenario (about 31 shots and 24 repetitions per session, roughly 6 minutes). An 8th file (user 243, `WarmUp` scenario, 350 shots) is a different drill and is left out of the cross-user comparisons - see Section 4.

0. Data loading & EDA

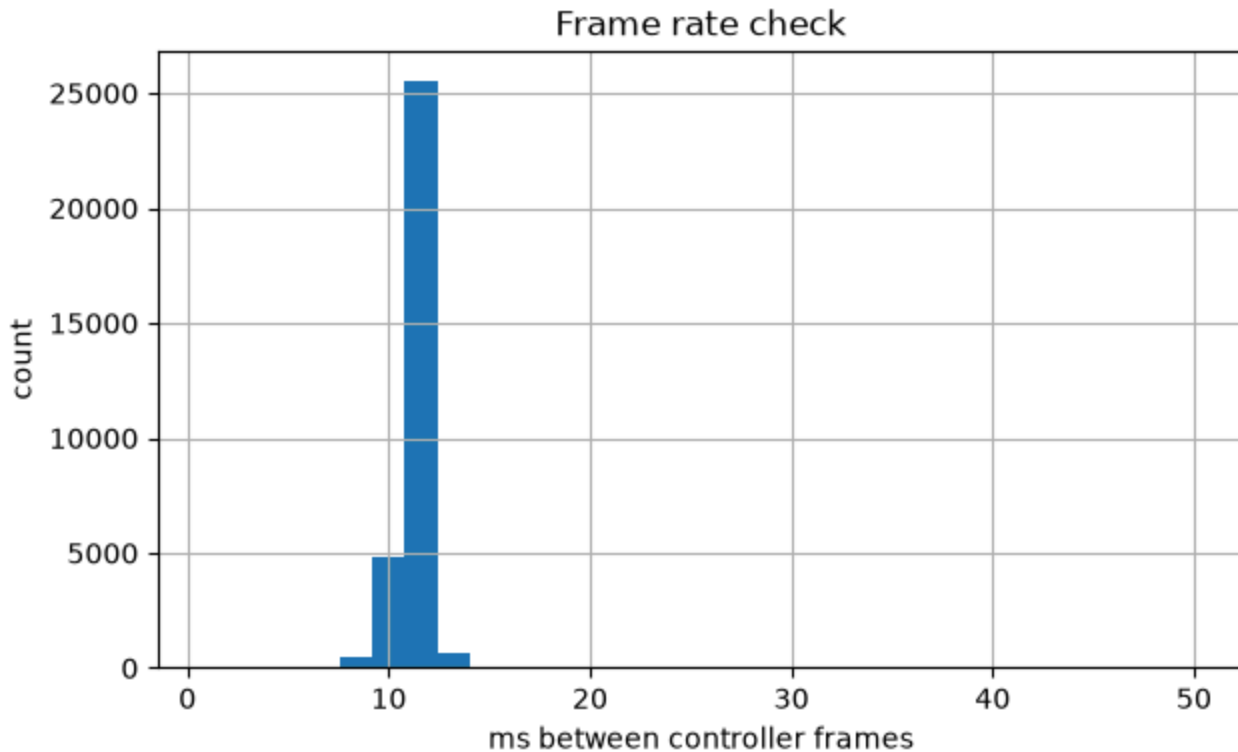
Filenames follow `Timestamp_UserId_SessionId_ScenarioID.event.json`. Each file is NDJSON (one JSON object per line), so we read it by streaming `json.loads` per line instead of `pandas.read_json` - files run 20-75MB, and loading the whole thing into memory at once buys nothing for a line-oriented format.

First, a look at one file's raw structure: event-type volumes and frame rate.

```
20260601072221_238_2566_11639.event.json
```

```
31735 controller
31735 headset
31735 eye_data
  144 drill_state_change
    96 shooting_requirement_change
    96 target_state_change
  48 buzzer
  31 shot_start
  31 shot_resolved
  24 repetition_start
  24 repetition_end
  1 session_metadata
  1 scenario_start
  1 location_metadata
  1 scenario_end
```

```
median frame gap: 11.0 ms (~91 Hz)
```



Timestamp note: `Time` is an integer millisecond epoch, but in this sample it decodes to a date around 2026 - almost certainly a synthetic or shifted clock rather than a real recording date. We only ever use *differences* between `Time` values (reaction time, window slicing), never the absolute wall-clock value, so this has no effect on any feature.

Event ordering: events of different `Type` are interleaved in the file and aren't guaranteed to be causally ordered even after sorting by `Time`. We found a concrete case of this: a `shooting_requirement_change` "clear" can land at the exact same millisecond as the `shot_start` that triggered it, and a plain stable sort leaves "clear" first - making it look like the requirement was already gone before the shot that satisfied it. `shooter/parse.py` breaks ties at equal timestamps by processing `shot_start` before `shooting_requirement_change`. After that fix, our recomputed valid-hit flag (`object_id` matched against whichever requirement was live at shot time) agrees with the engine's own `object_type` label on **100% of shots** across all 8 files - good evidence that the event model (drill -> repetition -> buzzer -> requirement -> shot) was reconstructed correctly.

disagreement between recomputed valid-hit and engine `object_type`: 0%

Eye-tracking data quality: `valid.<eye>` / `open.<eye>` record per-frame tracking confidence. One user (243) has consistently lower left-eye validity (about 75-78%) than right-eye (about 90-96%) across both of their sessions, despite `session_metadata` listing `dominant_eye = "Left"` - that pattern looks like a per-user calibration or sensor issue rather than ordinary blinking. `add_gaze_and_pupil` in `shooter/features.py` handles this by preferring the dominant eye but falling back to the other eye, per shot, whenever

the dominant eye doesn't clear a minimum valid-frame count in its window, instead of just returning NaN for that user's gaze features.

	user_id	dominant_eye	left_valid_rate	right_valid_rate
0	234	Right	0.99	0.98
1	235	Right	0.95	0.95
2	237	Right	0.80	0.89
3	238	Left	0.93	0.89
4	240	Left	0.94	0.93
5	244	Left	0.98	0.82
6	243	Left	0.78	0.90
7	243	Left	0.72	0.96

Building the per-shot feature table

The core join (`shooter/parse.py: build_shot_table`) replays every event in `Time` order, keeping drill, repetition, buzzer, and "live" shooting requirements as running state, and emits one row per `shot_resolved` joined to its matching `shot_start`, repetition, and active requirement. `shooter/features.py` then adds the feature columns on top of that table.

216 shots from 7 users (scenario `[np.int64(11634), np.int64(11635), np.int64(11639), np.int64(11640), np.int64(11641), np.int64(11642), np.int64(11646)]`; excluded scenarios: `{11601}`)

	user_id	rep_idx	reaction_time_s	radial_error_m	trigger_press_duration_s	trigger_sn
0	234	0	2.534	0.077496		0.623
1	234	1	2.379	0.021984		0.790
2	234	2	2.281	0.075626		0.489
3	234	3	1.945	0.125847		0.644
4	234	4	2.675	0.052760		0.044



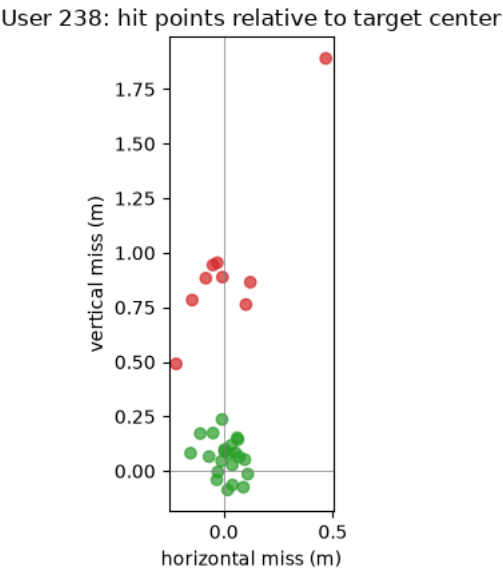
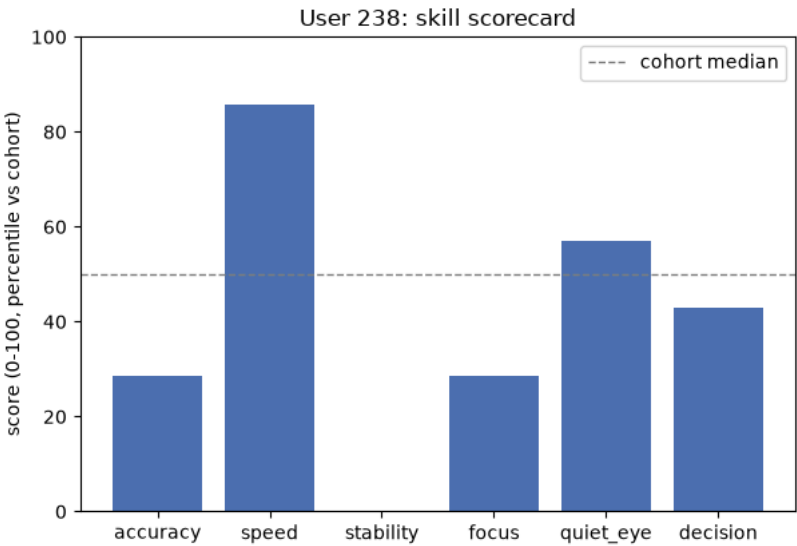
1. Shooter Dashboard

A plain-language summary for one shooter (user 238, chosen because their profile mixes clear strengths and clear weaknesses, which makes for a more useful example than someone who's simply good or bad across the board). Scores are **0-100 percentiles**

against the other 6 users in this cohort: 100 is best in the group, 50 is the middle. With only 7 users this is a coarse ranking, not a calibrated population norm (see Section 4).

	accuracy	speed	stability	focus	quiet_eye	decision
user_id						
234	0.19	2.18	0.03	1.78	0.80	0.74
235	0.17	2.36	0.02	1.65	0.86	0.84
237	0.40	2.11	0.05	5.24	0.47	0.71
238	0.36	2.00	0.08	5.75	0.47	0.71
240	0.39	2.12	0.06	9.99	0.64	0.71
243	0.34	2.40	0.06	7.55	0.06	0.60
244	0.22	2.50	0.04	5.36	0.45	0.84

	accuracy	speed	stability	focus	quiet_eye	decision
user_id						
234	71.0	43.0	71.0	71.0	86.0	71.0
235	86.0	29.0	86.0	86.0	100.0	93.0
237	0.0	71.0	43.0	57.0	43.0	43.0
238	29.0	86.0	0.0	29.0	57.0	43.0
240	14.0	57.0	14.0	0.0	71.0	43.0
243	43.0	14.0	29.0	14.0	14.0	14.0
244	57.0	0.0	57.0	43.0	29.0	93.0



Plain-language feedback for user 238: Your trigger finger is fast. You're the quickest in the group to fire once you're allowed to shoot (top score on speed). But that speed isn't paying off in steadiness: your trigger pull is the jerkiest in the group (lowest stability score), and your eyes drift the furthest from the target in the moment before you shoot (low focus score). Accuracy and decision-quality both sit below the group median. **Bottom line: you're shooting fast, but slow down on the final trigger squeeze and keep your eyes locked on the target through the shot - right now speed is costing you precision.**

2. Feature Explanation

The brief asks for 3-5 features, each from a different performance domain. Building the pipeline, we ended up with seven candidate metrics across five domains - more than one per domain in a couple of cases. Below is the **main set of five**, one per domain, followed by a derived sixth (decision quality) and then the two extra metrics we tested but didn't promote, along with the reasoning for leaving each one out.

Main features

#	Feature	Domain	What it measures	Data source
1	reaction_time_s	Timing / target acquisition	Seconds from buzzer(start) to shot_start	buzzer , shot_start
2	radial_error_m	Motor control / consistency	Distance from hit point to target center (x,y plane)	shot_resolved.hit_local_location
3	trigger_smoothness	Trigger control	Jaggedness of the final trigger pull before the shot	controller.trigger_pull_value
4	gaze_target_angle_deg	Eye behavior	Angle between gaze direction and target, averaged over the 500ms before the shot	eye_data
5	quiet_eye_duration_s	Cognitive	Duration of the final continuous on-target fixation right up to the trigger pull ("Quiet Eye")	eye_data

1. Reaction time (timing / target acquisition)

Time from the buzzer's "start" signal to the shot's trigger break - the most direct measure we have of how quickly a shooter engages a presented target, which is a tactical skill in its own right.

One assumption worth flagging: in multi-bullet repetitions, the second bullet's "reaction time" includes whatever time was already spent on the first bullet, so it reads as *time-to-fire-this-bullet-since-allowed* rather than a clean reflex measure. We're calling that out rather than treating the two cases as equivalent.

For product, this maps directly to a "draw speed" or "time-to-first-shot" metric for the shooter, and could back a pass/fail threshold in time-limited scenarios.

2. Precision and grouping (motor control / consistency)

Euclidean distance between the hit point and the target's center, in the plane perpendicular to the line of fire (x = lateral, y = vertical; we drop z , the depth into the target, since the engine already snaps it to the target plane and it carries no aim-quality signal here).

This is about as basic a shot-quality measure as exists: even a "valid" hit can land far from center, which matters for follow-up shot speed and lethality in a realistic scenario. Looking at `radial_error_m` across repeated shots at the same target row also gives a grouping/consistency read, not just a single-shot number.

For product, this maps straight onto the "shot grouping" visualizations shooters already expect from range training.

3. Trigger control

From the `trigger_pull_value` stream in the second before the shot, we take `trigger_smoothness`: the standard deviation of frame-to-frame change during the final press (lower = smoother). We're assuming the shot fires at or near the local max of the pull within that window.

A jerky or "slapped" trigger pull is a well-known cause of low-and-left misses in real shooting, and it's coachable independently of aim. A below-peers smoothness score is a concrete, drillable cue ("work on dry-fire trigger control") in a way that "shoot more accurately" isn't.

4. Gaze and target alignment (eye behavior)

Angle between the dominant eye's `world_gaze_direction` and the vector to the target center, averaged over the 500ms before the shot. This assumes gaze and target-center coordinates share the same world frame, which we checked by cross-referencing Section 0's hit-location coordinates against `location_metadata.target_base`.

"Look where you want to shoot" is foundational marksmanship coaching, and a large gaze-target angle right before the shot turns out to be a leading indicator of a miss (Section 3).

Of everything in this report, this is the strongest candidate for a real-time in-VR cue ("eyes on target"), since it precedes the shot rather than just describing it afterward.

5. Quiet Eye duration (cognitive)

How long, continuously, gaze stayed within 3° of the target right up to the trigger pull. We walk backward frame-by-frame from `shot_start` until gaze leaves that tolerance or a frame gap signals a blink or tracking dropout.

This is the "Quiet Eye" measure from sports-vision research (Vickers, 1996): the final fixation before a critical motor action, replicated specifically in marksmanship and police-shooting studies (Causer, Bennett & Williams, 2010), where a longer pre-shot Quiet Eye duration predicts a higher hit rate.

It's tempting to assume this just duplicates `gaze_target_angle_deg`, but the two ask different questions. The angle feature averages over a fixed 500ms window, so a shooter whose gaze wanders onto target for most of that window but breaks away right before firing still scores well on it. Quiet Eye asks specifically how long gaze was locked on, uninterrupted, through the moment of the shot - catching the failure mode the average can hide. A `0.0s` value means gaze was already off-target at the shot and is a real result, not missing data; `NaN` is reserved for windows with too few valid gaze frames to judge.

The 3° tolerance and the trigger-pull instant as the Quiet Eye "offset" are both literature-standard choices rather than something measured from this dataset - a tighter or looser tolerance would shift the absolute durations but not the cohort ranking (checked in Section 3). For product, this pairs with the gaze-angle cue as a single, well-validated number that's easy to coach directly - "hold your sight picture on target for at least half a second before you break the shot" is a known drill in real-range Quiet Eye training.

Bonus: decision quality (cognitive, derived)

Beyond the five main features, we also compute the share of a user's shots fired at a target that wasn't currently a "live" shooting requirement (for example, hitting a no-score or off-target object while the intended target was active). It's derived from the same event data as the others rather than measured directly off a sensor, which is why we're treating it as a bonus rather than one of the five.

It catches a failure mode none of the other features can see: "couldn't hit the right target" (a motor/aim problem) is different from "shot the wrong target" (a target-discrimination or impulse-control problem), and those call for different coaching entirely. A shooter with a high non-valid rate but otherwise good accuracy and reaction time probably needs discrimination drills, not marksmanship ones.

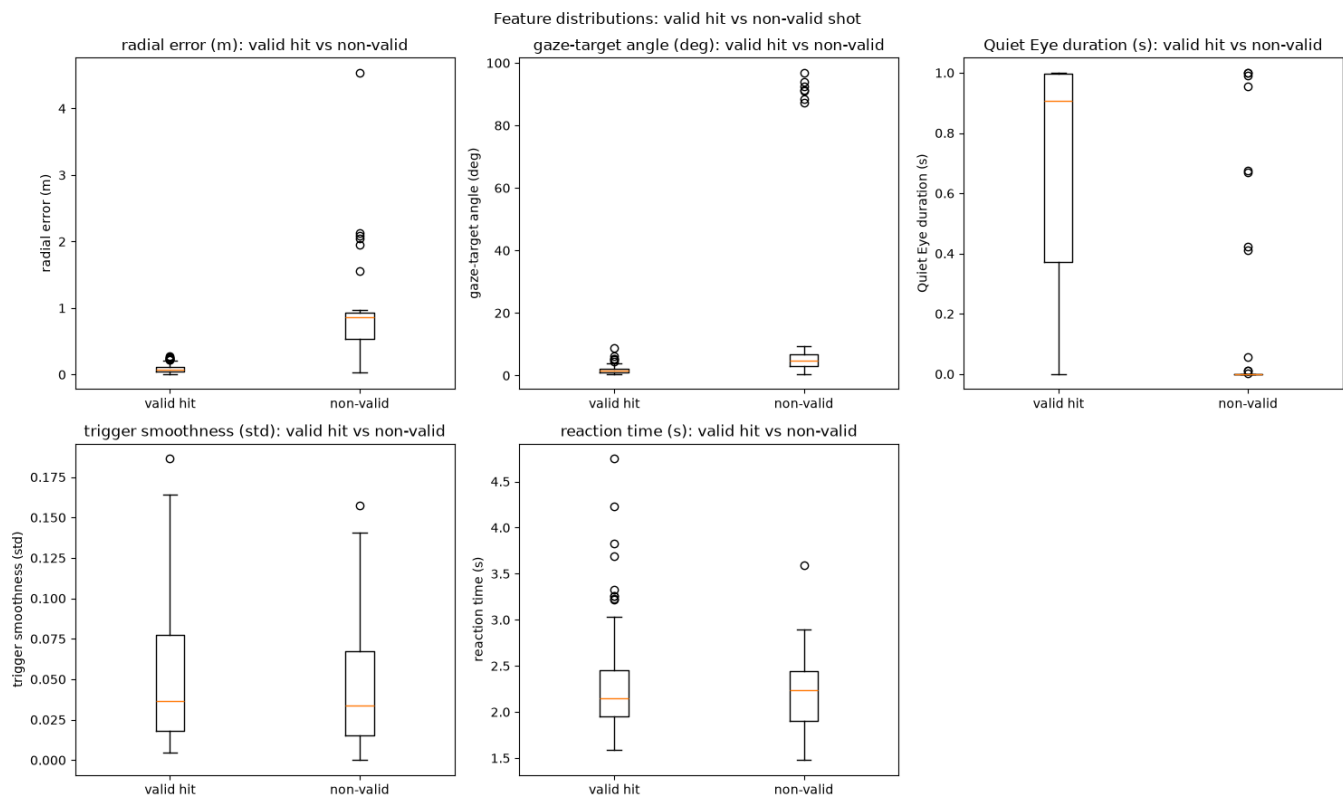
Supplementary features (tested, not in the main set)

Two more per-shot metrics came out of the same pipeline. Both are real, validated columns in `features.csv` - they just didn't make the headline five, for reasons specific to each:

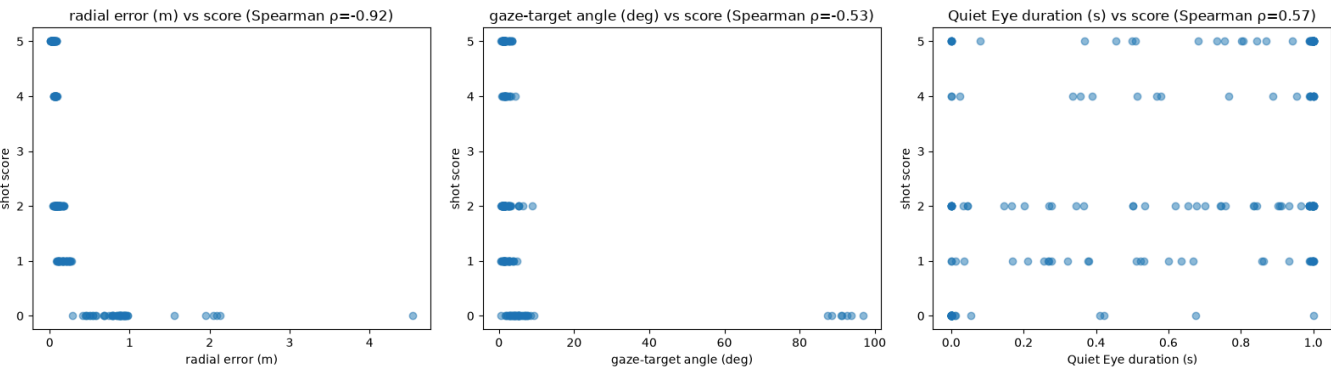
- **trigger_press_duration_s** (time from a half-way pull threshold to the shot) sits in the same domain as **trigger_smoothness** and is correlated with it. Smoothness is the more diagnostic and more obviously coachable of the two, so it became the main trigger-control feature, and duration is kept as a supporting number rather than giving trigger control two headline slots.
- **pupil_diameter_mm** (mean pupil size in the pre-shot gaze window, as a cognitive-load/arousal proxy) is conceptually interesting, but unlike every main feature it never showed a clean separation between valid hits and misses or a correlation with the engine's **score** field (Section 3 only runs that check on the main five). Without that evidence we're not comfortable presenting it as a validated shot-quality signal yet - flagged here as worth a second look once there's a clearer hypothesis for what pupil size should predict in this task. (**gaze_stability_deg**, the per-shot standard deviation of the gaze angle, is in the same boat: a reasonable idea, computed alongside the others, but not separately validated or promoted.)

3. Feature Validation / Supporting Analysis

The core trust check for every feature: does it actually separate good shots from bad ones?



Radial error and gaze-target angle both show a clear, large separation (misses have much larger radial error and much larger gaze-target angle): strong validation for both. Quiet Eye duration also separates the groups in the expected direction (valid hits have a longer final on-target fixation), though with more overlap than the angle feature, consistent with it capturing a related but distinct, noisier signal (a single critical instant versus an averaged window). Trigger smoothness and reaction time show weaker or no separation by hit/miss, which is expected: a shooter can have a clean, smooth, fast trigger pull and still miss for purely aim-related reasons. These two are technique measures, not outcome measures, and get validated differently below (against each other and against between-user variation, not against the binary hit/miss label).



All three correlate with the engine's own `score` field in the expected direction (more error or a wider gaze miss means a lower score; a longer Quiet Eye means a higher score), without us ever telling the pipeline what `score` was. That's independent validation that these features track real shot quality rather than an artifact of how they were built.

	reaction_time_s	radial_error_m	trigger_smoothness	gaze_targ
reaction_time_s	1.00	-0.11	-0.01	
radial_error_m	-0.11	1.00	0.05	
trigger_smoothness	-0.01	0.05	1.00	
gaze_target_angle_deg	0.00	0.55	-0.05	
quiet_eye_duration_s	0.01	-0.57	-0.12	
trigger_press_duration_s	0.34	-0.17	-0.09	
pupil_diameter_mm	0.23	-0.05	-0.23	



Most correlations are modest ($|p|$ mostly under 0.3), which suggests each feature is contributing something distinct rather than all of them re-measuring "had a good day." That's what the brief is after - different performance domains, not several views of the same underlying thing - and it holds for the two supplementary features too, not just the main five.

The one clear exception is `quiet_eye_duration_s` vs `gaze_target_angle_deg` ($\rho \approx -0.84$ per shot). Both measure gaze-target alignment from overlapping data, so a strong per-shot relationship isn't surprising. What actually matters is whether they *rank shooters* the same way - that's checked directly below.

per-user skill-score rank correlation (focus vs quiet_eye): 0.54

	focus	quiet_eye
user_id		
234	71.0	86.0
235	86.0	100.0
237	57.0	43.0
238	29.0	57.0
240	0.0	71.0
243	14.0	14.0
244	43.0	29.0

At $\rho \approx 0.5$, the per-user rankings agree only moderately, well below the per-shot correlation. User 240 is the clearest case: worst in the cohort on average gaze-target angle, but mid-pack on Quiet Eye duration. That's a shooter whose gaze wanders during the approach but snaps onto target right at the moment of the shot. A coaching report built on `gaze_target_angle_deg` alone would flag this person as "not looking at the target," but Quiet Eye duration tells a more precise story: eyes are on target when it counts, just not before that. That's the complementary information that justifies keeping both as main features instead of picking one.

	user_id	object_id	score	radial_error_m	reaction_time_s	trigger_smoothn
best shot	235	LINE_1/5/HEAD/A	5.0	0.007830	3.023	0.013
worst shot	237	WallBack_Collider	0.0	4.536881	2.240	0.069

The worst shot by radial error also has a far larger gaze-target angle than the best one. That's a single concrete case of the gaze feature explaining a miss that the precision number alone can't diagnose: was this a motor problem, or did the shooter just look at the wrong place?



4. Features Considered But Not Built

Beyond the two supplementary features in Section 2 (computed, just not promoted), a couple of other ideas didn't make it into the pipeline at all:

- **Headset/controller path stability before the shot** (hand/head sway). The frame data would support this, but conceptually it overlaps heavily with `trigger_smoothness`: both are really asking "how steady was the shooter right before the shot?" With only seven users, adding a third highly-correlated steadiness metric felt more likely to dilute interpretability than add signal. Worth revisiting if the cohort grows.
- **Convergence distance** (`eye_data.convergence`). It's in the schema, but was flat at 100.0 across the sample we explored, which suggests it isn't actively computed in this build or scenario - not reliable enough to build a feature on yet.

Valuable, but needs more data than this sample has:

- *Real per-user norms*. The 0-100 scores in this report are percentiles within a group of 7 people on one scenario - useful for comparing this specific cohort, but not a stable population baseline. Turning "70th percentile" into something that means the same thing for a new user next month would need norms built from dozens to hundreds of sessions per skill level.
- *Learning-curve features* - does reaction time drop across repetitions within a session, or across sessions for the same user? The sample is one session per user (243 has two), which isn't enough repeated sessions to fit a trend.

- *Recoil/follow-through and stance/posture features* would need sensor logs this schema doesn't have (full-body tracking, weapon recoil telemetry).
- *Audio* (verbal commands, ambient distraction) isn't referenced anywhere in the schema and would need an entirely new log type.

Logging changes that would help: an explicit `shot_break_time` captured directly off the trigger sensor (instead of inferring it from the `trigger_pull_value` curve) would make the trigger-control features exact rather than approximated. A per-frame `gaze_confidence` score, instead of the binary `valid / open` flags, would also remove the need to infer the eye-tracking fallback in Section 0 from raw validity rates.

reaction time range across cohort: 2.0s - 2.5s
 gaze-target angle range across cohort: 1.6deg - 10.0deg (6.1x spread)
 Quiet Eye duration range across cohort: 0.06s - 0.86s (14x spread)

5. Product Manager Summary

Scope: five features, one per domain (timing, motor control, trigger control, eye behavior, cognitive), plus a derived decision-quality metric. Two more metrics were built and validated alongside these but set aside as supplementary - see Section 2 for the reasoning on each.

What we learned about this cohort: reaction time is fairly uniform across the 7 shooters (roughly 2.0-2.5s); everyone engages at about the same speed once the buzzer goes. The real spread is in eye behavior. Gaze-target angle varies by roughly 6x between the most and least target-focused shooter, and Quiet Eye duration spreads even wider (see the numbers above). Eye behavior is the single most differentiating skill in this cohort, and it's also the domain that's easiest to turn into a real-time in-VR coaching cue: "look at the target," "hold your sight picture before you fire."

Strong and weak skills: decision quality (shooting the right target) and eye behavior (focus and Quiet Eye) separate the cohort the most; reaction time separates it the least, with trigger control and accuracy in between. Focus and Quiet Eye only agree moderately on how they rank shooters ($\rho \approx 0.5$, Section 3), so it's worth surfacing both rather than treating one as "the" eye metric - each catches a shooter the other would miss. One user, for example, ranks worst on average gaze alignment but mid-pack on Quiet Eye: eyes wander during the approach, but lock on right when it counts.

Most reliable features: `radial_error_m` and `gaze_target_angle_deg` are the two we'd trust most. Both show a large, sensible gap between valid hits and misses (Section 3), and both correlate with the engine's own shot score without ever having been told what that score was. `quiet_eye_duration_s` shows the same pattern, just more weakly, since it's a single-instant measure and noisier per shot - but it adds a distinct, literature-backed angle the others don't cover. `trigger_smoothness` and `reaction_time_s` are technique measures rather than outcome measures, so they validate differently (by between-user

spread, not hit/miss separation): trust them as *process* feedback, not as a stand-in for "did they hit."

What would improve logging or product feedback: capture an explicit trigger-break event instead of inferring it from the raw pull curve; log per-frame eye-tracking confidence instead of a binary valid/open flag, so gaze features don't need an inferred per-user eye fallback; and collect multiple sessions per user before showing anyone a percentile score, so "better than 70% of users" reflects a real population rather than these 7 people.